

Data Audit: Variable Types and Storage

The CONTENTS Procedure

Data Set Name	WORK.IMPORT	Observations	8000
Member Type	DATA	Variables	14
Engine	V9	Indexes	0
Created	05/02/2026 14:27:03	Observation Length	112
Last Modified	05/02/2026 14:27:03	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information	
Data Set Page Size	131072
Number of Data Set Pages	7
First Data Page	1
Max Obs per Page	1168
Obs in First Data Page	1135
Number of Data Set Repairs	0
Filename	/saswork/SAS_work7C59000033DF_odaws01-euw1.oda.sas.com/SAS_workDD89000033DF_odaws01-euw1.oda.sas.com/import.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	1610614249
Access Permission	rw-r--r--
Owner Name	u64499790
File Size	1MB
File Size (bytes)	1048576

Alphabetic List of Variables and Attributes					
#	Variable	Type	Len	Format	Informat
10	ad_clicked	Num	8	BEST12.	BEST32.
2	age	Num	8	BEST12.	BEST32.
12	avg_session_time	Num	8	BEST12.	BEST32.
13	bounce_rate	Num	8	BEST12.	BEST32.
8	cart_items	Num	8	BEST12.	BEST32.
4	device_type	Char	7	\$7.	\$7.
9	discount_seen	Num	8	BEST12.	BEST32.
3	gender	Char	6	\$6.	\$6.
6	pages_viewed	Num	8	BEST12.	BEST32.
7	previous_purchases	Num	8	BEST12.	BEST32.
14	purchase	Num	8	BEST12.	BEST32.
11	returning_user	Num	8	BEST12.	BEST32.
5	time_on_site	Num	8	BEST12.	BEST32.
1	user_id	Num	8	BEST12.	BEST32.

Data Audit: Missing Values (Numerical)

The MEANS Procedure

Variable	N	N Miss	Mean	Std Dev	Minimum	Maximum
user_id	7840	160	4003.14	2307.90	1.0000000	8000.00
age	7840	160	38.5957908	12.0339278	18.0000000	59.0000000
time_on_site	7840	160	15.7146735	8.3627638	1.0000000	30.0000000
pages_viewed	7840	160	9.9700255	5.4254854	1.0000000	19.0000000

Variable	N	N Miss	Mean	Std Dev	Minimum	Maximum
previous_purchases	7840	160	6.9641582	4.2811174	0	14.0000000
cart_items	7840	160	4.5302296	2.8658138	0	9.0000000
discount_seen	7840	160	0.5025510	0.5000254	0	1.0000000
ad_clicked	7840	160	0.5022959	0.5000266	0	1.0000000
returning_user	7840	160	0.5094388	0.4999428	0	1.0000000
avg_session_time	7840	160	15.7949987	9.8645556	0.5700000	44.7500000
bounce_rate	7840	160	49.9244018	29.1374486	0.0100000	99.9700000
purchase	7840	160	0.9281888	0.2581915	0	1.0000000

Data Audit: Frequency & Missingness (Categorical)

The FREQ Procedure

gender	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	160	2.00	160	2.00
Female	3909	48.86	4069	50.86
Male	3931	49.14	8000	100.00

device_type	Frequency	Percent	Cumulative Frequency	Cumulative Percent
	160	2.00	160	2.00
Desktop	2365	29.56	2525	31.56
Mobile	4685	58.56	7210	90.13
Tablet	790	9.88	8000	100.00

Data Audit: Extreme Observations & Outliers

The UNIVARIATE Procedure
Variable: age

Moments			
N	7840	Sum Weights	7840
Mean	38.5957908	Sum Observations	302591
Std Deviation	12.0339278	Variance	144.815418
Skewness	-0.0129995	Kurtosis	-1.1759005
Uncorrected SS	12813947	Corrected SS	1135208.06
Coeff Variation	31.1793787	Std Error Mean	0.13590936

Basic Statistical Measures			
Location		Variability	
Mean	38.59579	Std Deviation	12.03393
Median	39.00000	Variance	144.81542
Mode	43.00000	Range	41.00000
		Interquartile Range	21.00000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	283.9818	Pr > t	<.0001
Sign	M	3920	Pr >= M	<.0001
Signed Rank	S	15368360	Pr >= S	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	59
99%	59
95%	57

Quantiles (Definition 5)	
Level	Quantile
90%	55
75% Q3	49
50% Median	39
25% Q1	28
10%	22
5%	20
1%	18
0% Min	18

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
18	7951	59	7741
18	7824	59	7769
18	7766	59	7851
18	7758	59	7963
18	7721	59	7981

Missing Values			
Missing Value	Count	Percent Of	
		All Obs	Missing Obs
.	160	2.00	100.00

Data Audit: Extreme Observations & Outliers

The UNIVARIATE Procedure
Variable: time_on_site

Moments			
N	7840	Sum Weights	7840
Mean	15.7146735	Sum Observations	123203.04
Std Deviation	8.36276381	Variance	69.9358186
Skewness	-0.0245912	Kurtosis	-1.1876337
Uncorrected SS	2484322.43	Corrected SS	548226.882
Coeff Variation	53.2162748	Std Error Mean	0.09444779

Basic Statistical Measures			
Location		Variability	
Mean	15.71467	Std Deviation	8.36276
Median	15.74000	Variance	69.93582
Mode	12.21000	Range	29.00000
		Interquartile Range	14.36000

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	166.3848	Pr > t	<.0001
Sign	M	3920	Pr >= M	<.0001
Signed Rank	S	15368360	Pr >= S	<.0001

Quantiles (Definition 5)	
Level	Quantile
100% Max	30.00
99%	29.75
95%	28.67

Quantiles (Definition 5)	
Level	Quantile
90%	27.22
75% Q3	22.88
50% Median	15.74
25% Q1	8.52
10%	3.97
5%	2.57
1%	1.32
0% Min	1.00

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
1.00	7877	29.98	6550
1.00	3814	29.98	6570
1.01	7730	29.99	3702
1.01	4482	30.00	781
1.01	3801	30.00	2912

Missing Values			
Missing Value	Count	Percent Of	
		All Obs	Missing Obs
.	160	2.00	100.00

Data Audit: Extreme Observations & Outliers

The UNIVARIATE Procedure
Variable: bounce_rate

Moments			
N	7840	Sum Weights	7840
Mean	49.9244018	Sum Observations	391407.31
Std Deviation	29.1374486	Variance	848.99091
Skewness	0.01604017	Kurtosis	-1.2133657
Uncorrected SS	26196015.6	Corrected SS	6655239.74
Coeff Variation	58.3631402	Std Error Mean	0.32907394

Basic Statistical Measures			
Location		Variability	
Mean	49.92440	Std Deviation	29.13745
Median	49.50000	Variance	848.99091
Mode	53.34000	Range	99.96000
		Interquartile Range	51.31500

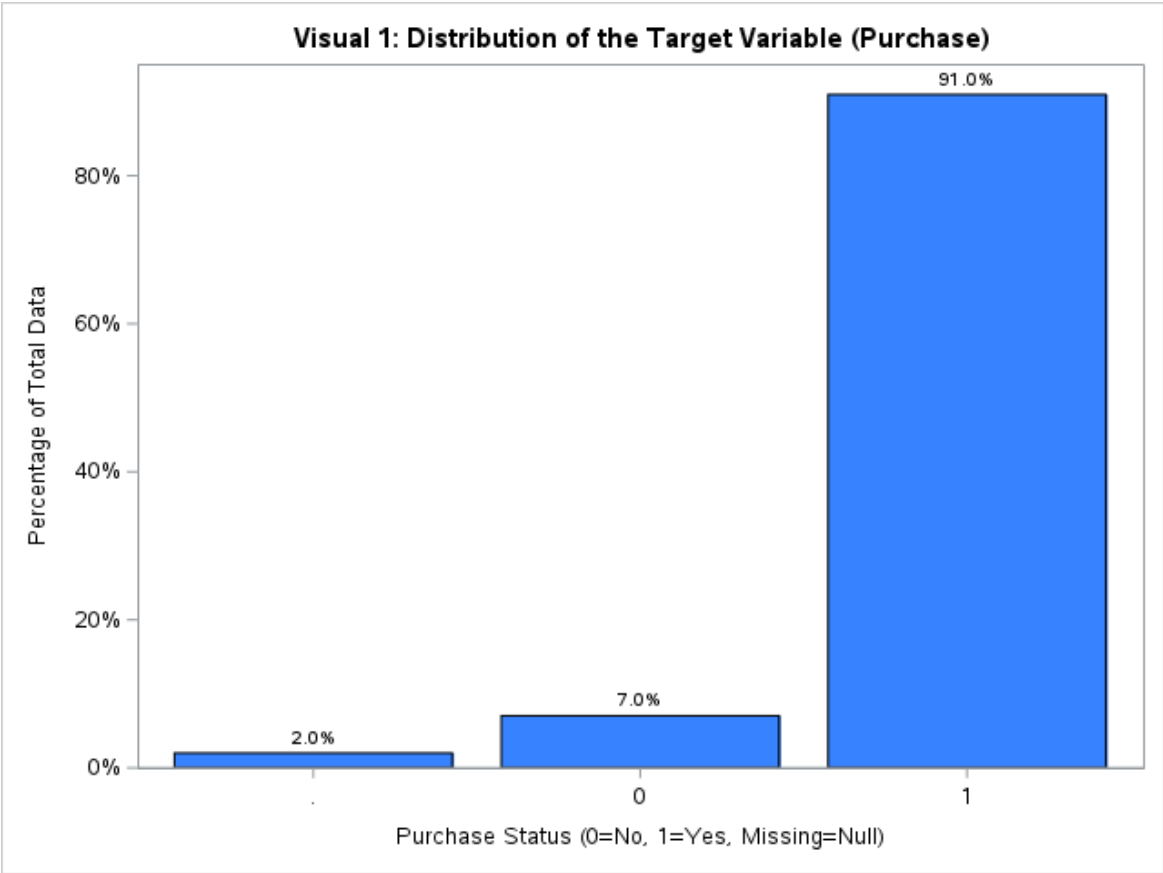
Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	151.7118	Pr > t	<.0001
Sign	M	3920	Pr >= M	<.0001
Signed Rank	S	15368360	Pr >= S	<.0001

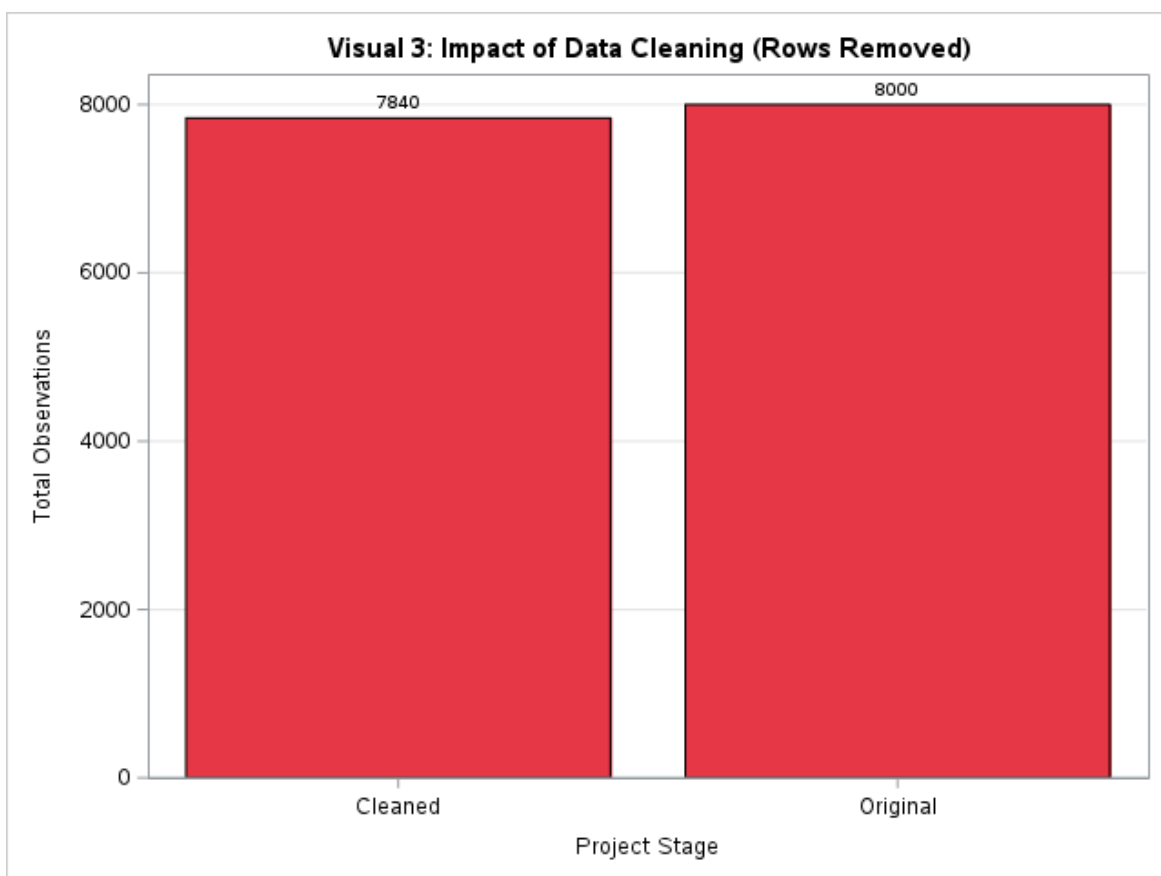
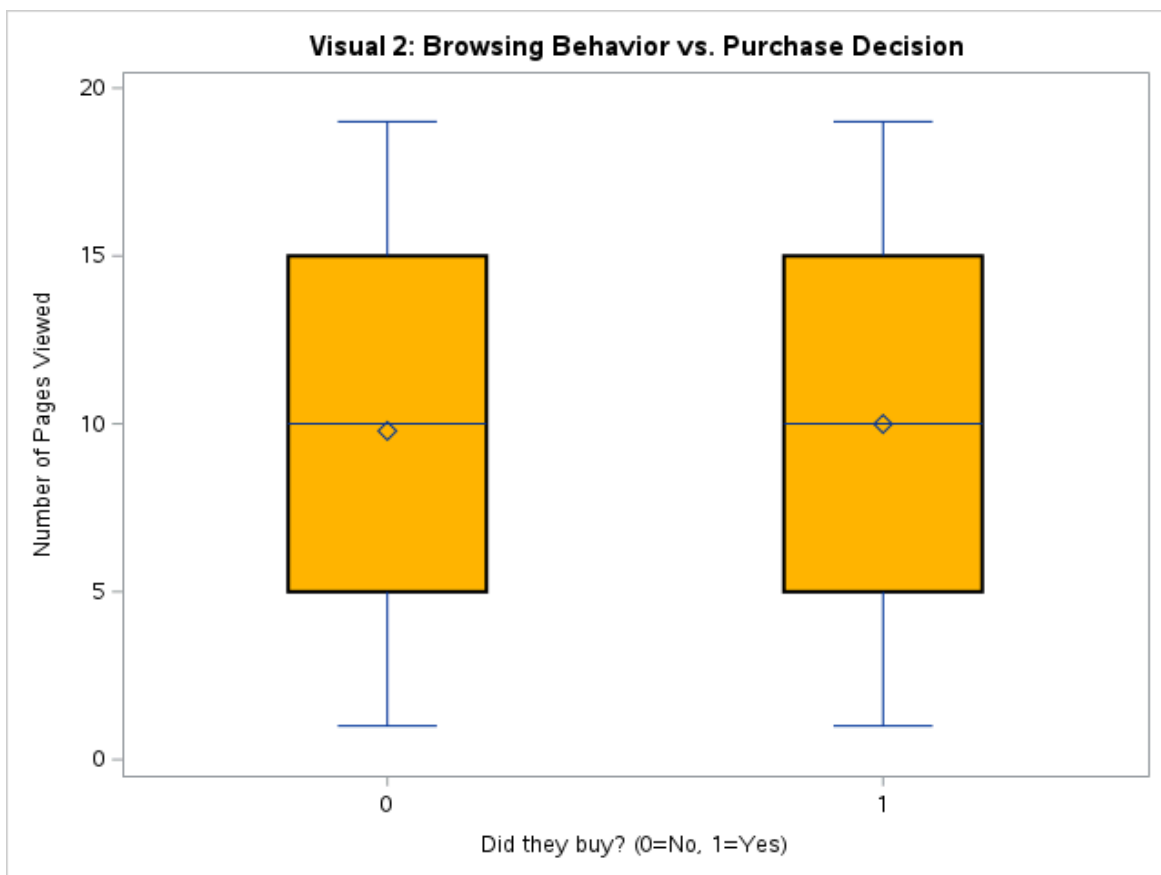
Quantiles (Definition 5)	
Level	Quantile
100% Max	99.970
99%	98.960
95%	95.255

Quantiles (Definition 5)	
Level	Quantile
90%	90.365
75% Q3	75.990
50% Median	49.500
25% Q1	24.675
10%	9.605
5%	4.845
1%	0.940
0% Min	0.010

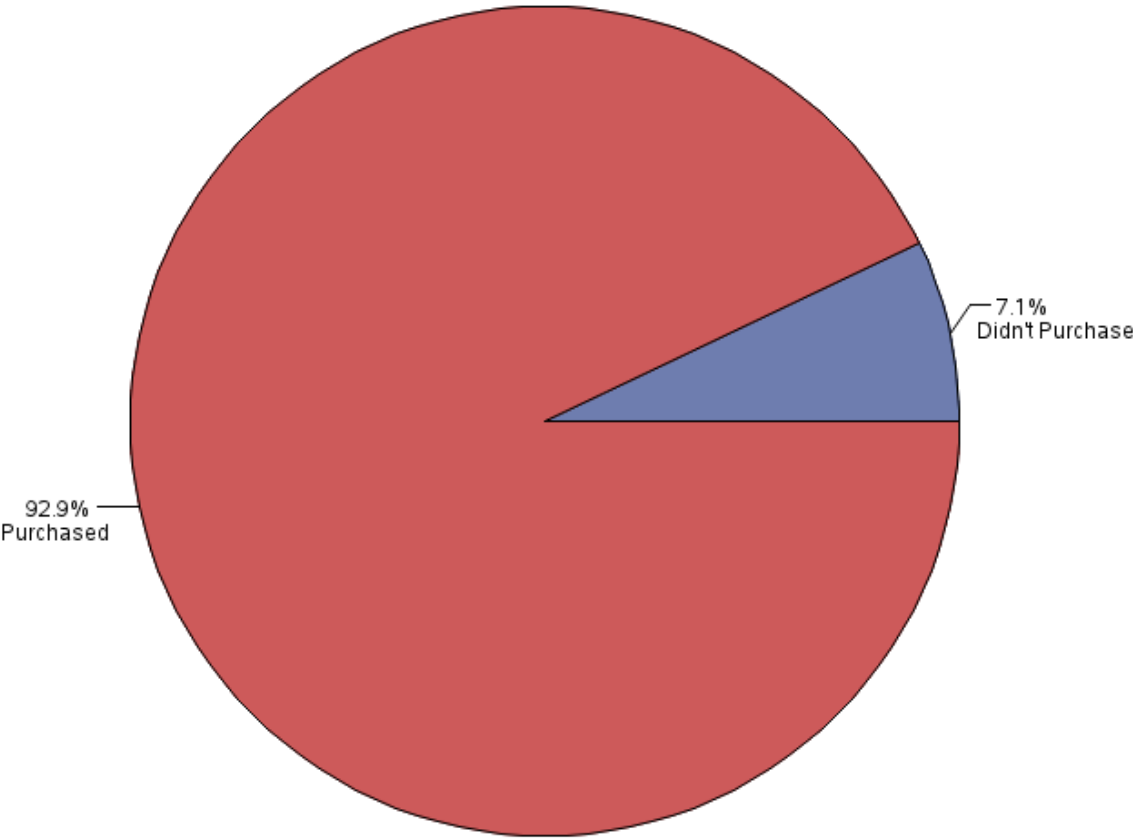
Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
0.01	7508	99.95	1514
0.01	4145	99.95	2087
0.01	3584	99.96	1806
0.02	445	99.97	3486
0.03	857	99.97	5650

Missing Values			
Missing Value	Count	Percent Of	
		All Obs	Missing Obs
.	160	2.00	100.00



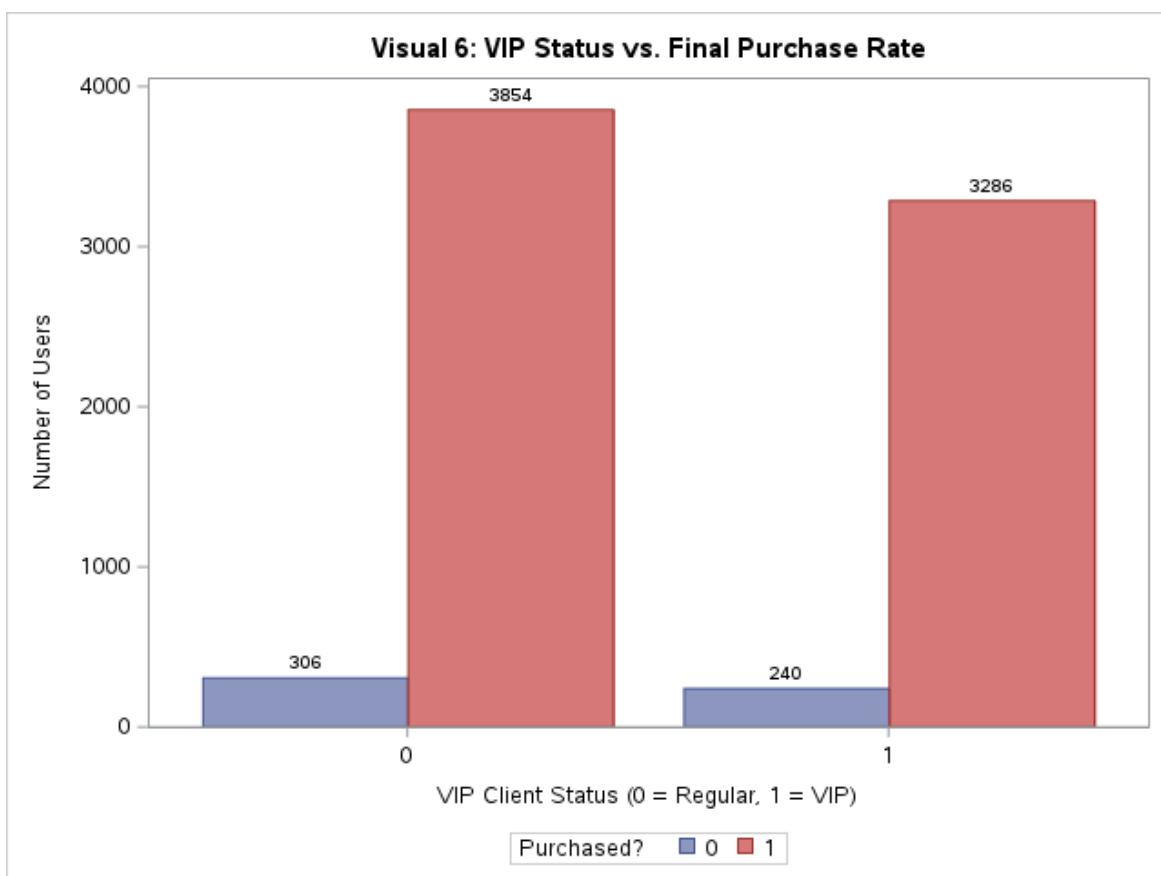
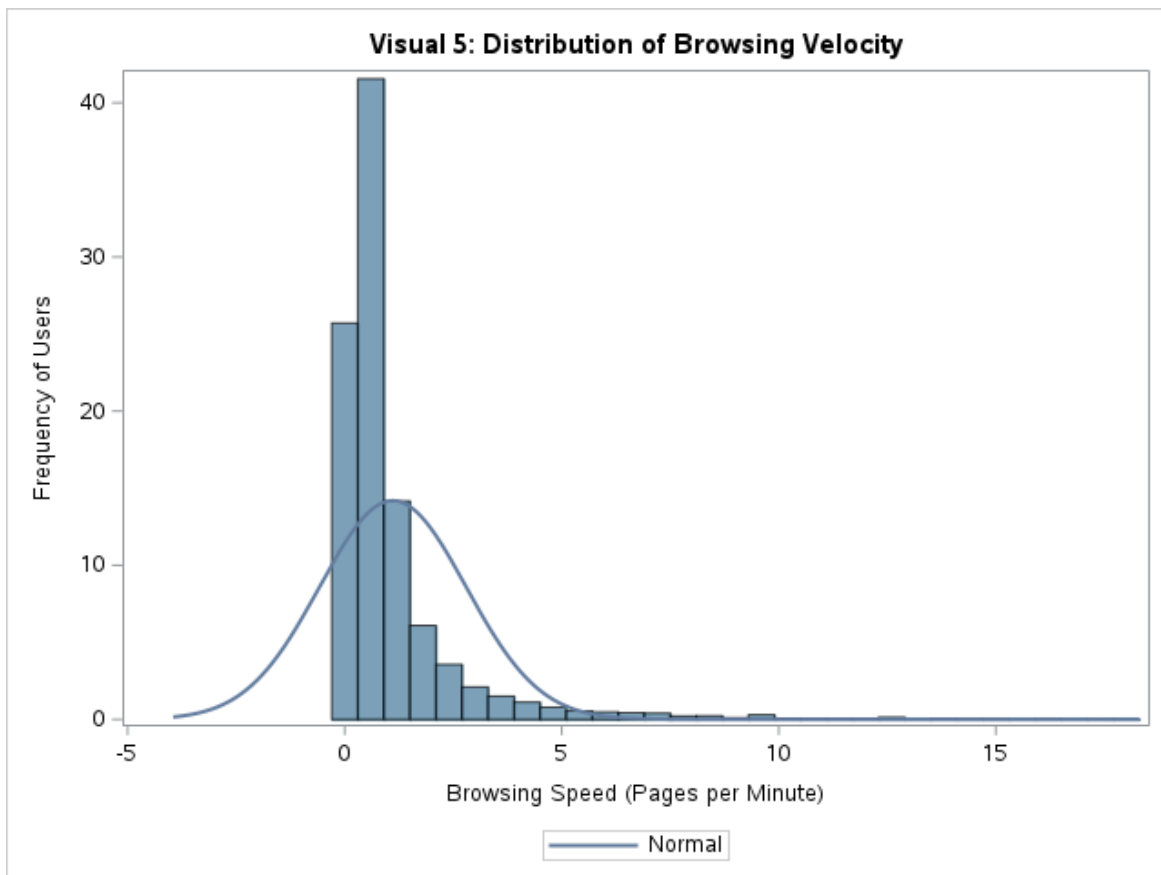


Visual 4: Purchase Distribution Analysis
PERCENT of purchase



Feature Engineering Audit: The Big Four Model Inputs

Obs	user_id	velocity	cart_density	vip_client	time_per_item
1	1	0.62016	0.12500	0	12.9000
2	2	0.57582	0.66667	1	2.6050
3	3	1.03093	0.00000	1	0.0000
4	4	0.22017	0.20000	1	22.7100
5	5	0.34156	0.44444	0	6.5875
6	6	0.03759	7.00000	0	3.8000
7	7	1.44737	0.54545	1	1.2667
8	8	0.25945	0.28571	0	13.4900
9	9	1.28440	0.64286	1	1.2111
10	10	0.30090	2.66667	0	1.2463



Binary Logistic Regression Model

The SURVEYSELECT Procedure

Selection Method	Simple Random Sampling
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Input Data Set	ECOM_FEATURES
Random Number Seed	12345
Sampling Rate	0.7
Sample Size	5381
Selection Probability	0.700104
Sampling Weight	0
Output Data Set	ECOM_SPLIT

Binary Logistic Regression: Predicting Purchase Intent

The LOGISTIC Procedure

Model Information	
Data Set	WORK.ECOM_SPLIT
Response Variable	purchase
Number of Response Levels	2
Model	binary logit
Optimization Technique	Fisher's scoring

Number of Observations Read	5381
Number of Observations Used	5381

Response Profile		
Ordered Value	purchase	Total Frequency
1	1	5013
2	0	368

Probability modeled is purchase='1'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	2686.593	2576.280
SC	2693.184	2609.233
-2 Log L	2684.593	2566.280

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	118.3131	4	<.0001
Score	111.2757	4	<.0001
Wald	97.7771	4	<.0001

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq	Standardized Estimate
Intercept	1	2.4811	0.1194	431.7866	<.0001	
velocity	1	-0.0169	0.0333	0.2585	0.6112	-0.0157
cart_density	1	0.6303	0.1174	28.8456	<.0001	0.4363
vip_client	1	0.1748	0.1106	2.4973	0.1140	0.0480
time_per_item	1	-0.0525	0.00797	43.3117	<.0001	-0.1538

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
velocity	0.983	0.921	1.050
cart_density	1.878	1.492	2.364
vip_client	1.191	0.959	1.479
time_per_item	0.949	0.934	0.964

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	70.2	Somers' D	0.404
Percent Discordant	29.8	Gamma	0.404
Percent Tied	0.0	Tau-a	0.052
Pairs	1844784	c	0.702

Model Validation: Full Accuracy & Prediction Audit

The FREQ Procedure

Frequency	Table of purchase by I_purchase		
	purchase	I_purchase(Into: purchase)	
		Purchased	Total
	Didn't Purchase	178	178
	Purchased	2127	2127
	Total	2305	2305

Executive Summary: Accuracy, Precision, and Recall

The LOGISTIC Procedure

Classification Table									
Prob Level	Correct		Incorrect		Percentages				
	Event	Non-Event	Event	Non-Event	Correct	Sensitivity	Specificity	Pos Pred	Neg Pred
0.500	2123	0	178	4	92.1	99.8	0.0	92.3	0.0

Getting Dataset Ready for visuals

The LOGISTIC Procedure

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vip_client	1	0.1748	0.1106	2.4973	0.1140	0.0480
time_per_item	1	-0.0525	0.00797	43.3117	<.0001	-0.1538

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